Hyperspectral Image Classification Powered by Khatri-Rao Decomposition-Based Multinomial Logistic Regression

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***Abstract*— Multinomial logistic regression (MLR) is of great significance in hyperspectral image (HSI) classification within remote sensing community. It seeks the optimal regressors with logistic loss function. In the past, spatial information has been widely used to improve its classification performance by means of some prepared feature or postprocessing techniques. To better understand HSI data and learn more effective spatial feature, in this article, a joint optimization framework that combines MLR classifier training with Khatri-Rao decomposition-based feature learning is proposed for HSI classification. It is called Khatri-Rao decomposition-based MLR (KR-MLR) algorithm. With Gabor feature as an input, KR-MLR customizes two data-oriented strategies. One is to insert a feature learning layer after the initial input and optimize it with classifier concurrently. Moreover, Khatri-Rao decomposition is utilized to convert the problem into tensor space and make it feasible in computation. Another is to add local regularization term to preserve discriminant information as well. The regressors and feature factor matrices are optimized in iterative fashion. The proposed KR-MLR is investigated on four popular HSI datasets. The experimental results show that KR-MLR outperforms other prior arts, proving it a competitive and promising classifier.**

***Index Terms*— Gabor wavelet, hyperspectral image (HSI) classification, Khatri-Rao decomposition, multinomial logistic regression (MLR), tensor analysis.**

I. Introduction

With the recent development of remote sensing devices, it is getting easier to acquire hyperspectral image (HSI) data [1], [2]. A typical HSI has hundreds or even thousands of narrow spectral bands, which cover a wide spectrum range from ultraviolet, visible, near infrared, to mid infrared. Its inherent abundant information facilitates numerous applications, such as environment monitoring, agriculture, and mineralogy [3]–[6]. An indispensable technique in these use cases is image classification. In the past decades, it has become pretty active and fostered lots of classifiers [7]–[11]. A popular one of them is multinomial logistic regression (MLR) [12]. It receives the name for the utilized logistic loss function. The literature witnesses its success. Li *et al.* [12] introduce sparse MLR to control classifier complexity and enhance its generalization capacity. Li *et al.* [13] propose semisupervised MLR with the aid of Laplacian regularization to handle unlabeled training samples. Li *et al.* [14] present subspace MLR to deal with noisy and highly mixed pixels in Bayesian framework. Qian *et al.* [15] propose structured sparse logistic regression model coupling with 3-D discrete wavelet transform features. In the recent background of deep learning, various deep neural network architectures are customized for HSI classification as well, such as spectral-spatial residual network (SSRN) [16], fast 3-D convolutional neural network (fast-3D-CNN) [17], recursive autoencoders [18], and multitask learning [19]. The current HSI classification works also include label similarity probability filter [20] and sparse representation-based ones [21]. More applications for MLR and neural network especially deep learning are reported in literature [22]–[24], [25], [26]–[29] beyond HSI classification. All prior arts show excellent results.

Previous works have proved that spatial information offers an effective means to boost classification accuracy [30]–[34]. That is because the neighboring pixels contain similar materials and thus share similar electromagnetic characteristics. Generally, spatial information can be utilized in three occasions, preclassification stage, classification stage, and postclassification stage.

A. Preclassification Stage

The typical methods are to extract spatial feature before classification [35]. As a benchmark, Gabor feature is generated by a filter bank covering various scales and orientations [33], [36]. The redundancy of filters brings about powerful feature as well as huge dimensionality. A compromise is to calculate it on a few top principal components (PCs). The other popular ones also include 3-D discrete wavelet feature [15], multiattribute profile feature [34], [37], and invariant attribute profiles (IAPs) feature [38].

B. Classification Stage

It learns spatial feature or discovers spatial structure during classifier training. The classical methods include neural network models and tensor models. For the former, they learn spatial feature by convolutional layer and pooling layer [16]–[18]. For the latter, they learn spatial structure by various tensor decomposition methods. A popular one is Khatri-Rao decomposition, which is also known as CANDECOMP/PARAFAC (CP) decomposition [39], [40].

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C. Postclassification Stage

A common strategy is to use some image segmentation techniques to create a smoother map based on the neighborhood relationship and pixelwise classification result. The related techniques involve minimum spanning forest [30], watershed [41], multilevel logistic prior [12], loopy belief propagation [42], and so on.

The spatial feature extraction techniques (preclassification stage) have advantages of high accuracy and low computation cost. They take one or two order of magnitude less time at least than deep algorithms (classification stage) especially on large scale datasets and provide higher accuracy than tensor techniques. At the same time, they are more stable than postprocessing (postclassification stage). However, spatial features still have some disadvantages. Take Gabor feature as an example. It suffers from two flaws.

1) The feature extraction process is independent of the classifier, and the feature may be not optimal in the sense of classification.

2) The feature extraction process rarely involves class information. All the efforts are to be devoted to resolve both issues.

In this article, a joint optimization framework that couples MLR classifier training with spatial feature learning is proposed for HSI classification. It aims to seek the most discriminant feature without significantly increasing the computation cost. It is implemented by Khatri-Rao decomposition and thus called Khatri-Rao decomposition-based MLR (KR-MLR) algorithm. With Gabor feature as an input, KR-MLR customizes two data-oriented strategies. To handle the first issue, KR-MLR inserts a feature learning layer after the initial input and optimizes it with classifier concurrently. Moreover, Khatri-Rao decomposition is utilized to convert the problem into tensor space and make it feasible in computation. It is prone to provide more effective feature, which facilitates classification. To handle the second issue, KR-MLR introduces a discriminant Laplacian regularization for the learned factor matrices to preserve class information to the most extent. The regressor and feature factor matrices are iteratively optimized by solving a series of monotonic and lower bounded subproblems. The advantages of KR-MLR are threefolds.

1) It provides an all-in-one optimization objective and solution for classifier training and feature learning, which creates combined force.

2) Tensor analysis and Khatri-Rao decomposition techniques greatly reduce the computation cost and ensure the availability of KR-MLR.

3) Laplacian regularization enhances the discriminant ability. Experiments on several HSI datasets validate its effectiveness.

The rest of this article is organized as follows. In Section II, MLR and Gabor filter-related techniques are briefly reviewed. Section III details the proposed KR-MLR model in tensor form with the aid of Khatri-Rao decomposition first and then focuses on its optimization procedure. Section IV illustrates its effectiveness by classification experiments on several popular HSI datasets. Finally, Section V summaries this article with some remarks.

II. MLR-Related Work

A. MLR

Let $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n) \in \mathbb{R}^{d\times n}$ be a training set of the investigated HSI data, where d and n are the number of spectral bands and labeled pixels, respectively. Let m be the class number. For each $\mathbf{x}_i$, its class label is written as $y_i \in \{1, \ldots, m\}$. MLR defines the posterior distribution as follows [13]:

$$p(y_i = k|\mathbf{x}_i, \mathbf{w}) = \frac{\exp\left[\mathbf{w}^{(k)^T}\mathbf{x}_i\right]}{\sum_{k=1}^{m}\exp\left[\mathbf{w}^{(k)^T}\mathbf{x}_i\right]} \tag{1}$$

where $\mathbf{w} = [\mathbf{w}^{(1)}; \mathbf{w}^{(2)}; \ldots; \mathbf{w}^{(m)}]$ are the regressors, which determine the separating hyperplanes. To avoid the underlying ill-conditioning problem, $\mathbf{w}$ is usually characterized by some priors. The most common one is Gaussian density [14], i.e., $p(\mathbf{w}) \propto \exp(-(\lambda/2)\|\mathbf{w}\|_2^2)$. Following the maximum *a posteriori* (MAP) estimation, the optimization objective can be eventually formulated as [13]:

$$\mathbf{w}_{\text{MAP}} = \underset{\mathbf{w}}{\text{argmax}} l(\mathbf{w}) + \log p(\mathbf{w}) \tag{2}$$

where $l(\mathbf{w}) = \log\prod_{i=1}^{n} p(y_i|\mathbf{x}_i, \mathbf{w})$ is the log-likelihood function.

B. Gabor Filter

Gabor filters have been widely used in texture analysis, image classification, and biometric identification [36], [43], [44]. It is defined by a harmonic function multiplied by a Gaussian function. A typical 2-D Gabor filter can be written as the following equation under coordinate (u, v):

$$h(u, v, s, \theta, \gamma, \eta, f) = \frac{f'^2}{\pi\gamma\eta}\exp\left(-\left(\frac{f'}{\gamma}\right)^2 u'^2 - \left(\frac{f'}{\eta}\right)^2 v'^2\right) \times \exp\left(j\left(2\pi f' u' + \frac{\pi}{2}\right)\right) \tag{3}$$

where

$$u' = u\cos\theta + v\sin\theta \tag{4}$$

$$v' = -u\sin\theta + v\cos\theta \tag{5}$$

$$f' = \frac{f}{2^{\frac{s-1}{2}}} \tag{6}$$

where s represents the scale. If the number of scales is set to S, then s covers the whole integer region $[1, S]$, i.e., $s \in \{1, 2, \ldots, S\}$. θ represents the orientation. Given an orientation number Θ, the values for θ are generated by dividing π into Θ equal parts. That means $\theta \in \{0, (1/\Theta)\pi, (2/\Theta)\pi, \ldots, ((\Theta - 1)/\Theta)\pi\}$. γ, η, and f are used to construct the standard derivation of the Gaussian function. With the variations of s and θ, it can produce lots of filters. To control the feature dimensionality, Gabor feature is usually generated on the top PCs [33]. It should be noted that the Gabor filter in (3) is a complex function. After convolving the filter bank, the absolute values are used as feature output.

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III. Khatri-Rao Decomposition-Based Multinomial Logistic Regression

A. Definitions and Notations

For the sake of convenience, before formally discussing the proposed classifier model KR-MLR, some related definitions and notations used in this article are presented at first.

Definition 1 (n-Mode Flattening): Let $\mathcal{A} \in \mathbb{R}^{d_1 \times d_2 \times \cdots d_N}$ be an N-order tensor where d_n is the dimensionality along nth axis. The n-mode flattening matrix is composed by a set of d_n-vectors, each of which is generated by traversing nth axis while keeping the other indexes fixed. It can be denoted by

$$\mathbf{A}_{(n)} \in \mathbb{R}^{d_n \times (d_1 \times d_2 \cdots \times d_{n-1} \times d_{n+1} \cdots \times d_N)}. \tag{7}$$

Definition 2 (n-Mode Product): The n-mode product of a tensor $\mathcal{A} \in \mathbb{R}^{d_1 \times \cdots \times d_n \times \cdots d_N}$ by a matrix $\mathbf{B} \in \mathbb{R}^{r \times d_n}$ is defined by

$$\mathcal{C} = \mathcal{A} \times_n \mathbf{B} \in \mathbb{R}^{d_1 \times \cdots \times r \times \cdots d_N} \tag{8}$$

where each entry in $\mathcal{C}$ is calculated by

$$c_{i_1,\ldots,i_{n-1} j i_{n+1},\ldots,i_N} = \sum_{i_n=1}^{d_n} a_{i_1,\ldots,i_{n-1} i_n i_{n+1},\ldots,i_N} b_{j i_n} \tag{9}$$

where $1 \le i_k \le d_k$ and $1 \le j \le r$. $a, b,$ and c with subscripts are the elements of $\mathcal{A}, \mathbf{B},$ and $\mathcal{C}$, respectively. The n-mode product can be written in n-mode flattening as well

$$\mathbf{C}_{(n)} = \mathbf{B}\mathbf{A}_{(n)}. \tag{10}$$

Definition 3 (Khatri-Rao Product): The Khatri-Rao product of matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_r] \in \mathbb{R}^{m \times r}$ and $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r] \in \mathbb{R}^{n \times r}$ is defined by

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{b}_1 & a_{21}\mathbf{b}_2 & \cdots & a_{r1}\mathbf{b}_r \\ a_{12}\mathbf{b}_1 & a_{22}\mathbf{b}_2 & \cdots & a_{r2}\mathbf{b}_r \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m}\mathbf{b}_1 & a_{2m}\mathbf{b}_2 & \cdots & a_{rm}\mathbf{b}_r \end{bmatrix} \in \mathbb{R}^{(mn) \times r} \tag{11}$$

where a_{ij} is the jth entry of $\mathbf{a}_i$.

Definition 4 (Kronecker Product): The Kronecker product of matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_r] \in \mathbb{R}^{m \times r}$ and $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_s] \in \mathbb{R}^{n \times s}$ is defined by

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{21}\mathbf{B} & \cdots & a_{r1}\mathbf{B} \\ a_{12}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{r2}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m}\mathbf{B} & a_{2m}\mathbf{B} & \cdots & a_{rm}\mathbf{B} \end{bmatrix} \in \mathbb{R}^{(mn) \times (rs)} \tag{12}$$

where a_{ij} is the jth entry of $\mathbf{a}_i$.

B. Proposed KR-MLR

As mentioned before, spatial feature extraction techniques have advantages of high accuracy and low computation cost, but they suffer from two flaws.

1) Their feature extraction processes are independent of the classifier and the features may be not optimal in the sense of classification.

2) Some techniques such as Gabor feature rarely involve class information.

To overcome both issues and better understand the HSI input, a unified optimization framework that couples feature learning with MLR classifier training is proposed. It is implemented by Khatri-Rao decomposition in tensor space and named KR-MLR. It aims to provide an all-in-one solution to learn more effective spatial feature along with classifier.

It is based on Gabor feature [33]. Let d_1 be the number of extracted PCs. Let d_2 and d_3 be the number of filter orientations and scales. Thus, the resulted Gabor feature can be stacked by traversing all filters as $\mathbf{x}_i \in \mathbb{R}^d$, where $d = d_1 \times d_2 \times d_3$. For feature learning, an intuitive vector-based linear mapping function $f : \mathbf{x}_i \to \mathbf{A}^{\mathbf{T}}\mathbf{x}_i$ can be a candidate, where $\mathbf{A} \in \mathbb{R}^{d_1 d_2 d_3 \times r}$ is the learning factor and it represents $\mathbf{x}_i$ in more discriminant form. With joint optimization in mind, a new MLR model is proposed as follows:

$$p(y_i = k|\mathbf{x}_i, \mathbf{w}) = \frac{\exp\left[\mathbf{w}^{(k)^T}\mathbf{A}^T\mathbf{x}_i\right]}{\sum_{k=1}^{m}\exp\left[\mathbf{w}^{(k)^T}\mathbf{A}^T\mathbf{x}_i\right]}. \tag{13}$$

From the aspect of function, (13) has overcome the first disadvantage of traditional Gabor feature and realized feature learning under MLR framework. However, it is infeasible in computation. Specifically, it involves a $d_1d_2d_3r$-by-$d_1d_2d_3r$ (nearly 100 000-by-100 000) matrix inversion. It is unbearable in computation cost, let alone the whole solving process. Even if a powerful workstation is utilized, the whole process still does not finish yet after a week.

The immediate problem is to reduce the computation complexity of (13). That means the number of unknown elements in $\mathbf{A}$ should be reduced as much as possible. To this end, an alternative is to break $\mathbf{A}$ into smaller ones. Specifically, Khatri-Rao decomposition can be used as a candidate solution. Suppose that $\mathbf{A}$ is approximated by its Khatri-Rao decomposition as $\mathbf{A} \approx \mathbf{D}\odot\mathbf{C}\odot\mathbf{B}$, where $\mathbf{B} \in \mathbb{R}^{d_1 \times r}$, $\mathbf{C} \in \mathbb{R}^{d_2 \times r}$, and $\mathbf{D} \in \mathbb{R}^{d_3 \times r}$, then KR-MLR can be eventually written in tensor space as follows:

$$\begin{aligned} p(y_i = k|\mathcal{X}_i, \mathbf{w}) &= \frac{\exp\left[\mathbf{w}^{(k)^T}(\mathbf{D}\odot\mathbf{C}\odot\mathbf{B})^T\mathrm{vec}(\mathcal{X}_i)\right]}{\sum_{k=1}^{m}\exp\left[\mathbf{w}^{(k)^T}(\mathbf{D}\odot\mathbf{C}\odot\mathbf{B})^T\mathrm{vec}(\mathcal{X}_i)\right]} \\ &= \frac{\exp\left[\mathbf{w}^{(k)^T}\mathrm{diag}(\mathcal{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T)\right]}{\sum_{k=1}^{m}\exp\left[\mathbf{w}^{(k)^T}\mathrm{diag}(\mathcal{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T)\right]} \\ &= \frac{\exp\left[\mathbf{w}^{(k)^T}\mathrm{diag}(\mathbf{B}^T\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C}))\right]}{\sum_{k=1}^{m}\exp\left[\mathbf{w}^{(k)^T}\mathrm{diag}(\mathbf{B}^T\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C}))\right]} \end{aligned} \tag{14}$$

where $\mathcal{X}_i \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ is the tensor form of $\mathbf{x}_i$. The second identical deformation depicts the 1-mode flattening form for solving $\mathbf{B}$. For $\mathbf{C}$ and $\mathbf{D}$, it can be rewritten in the corresponding flattening forms as well. diag$(\cdot)$ is defined as follows. If the input is a vector, it constructs a diagonal matrix; if the input is a square matrix, it outputs the diagonal entries as a column vector; if the input is a three-order tensor, it outputs a column vector as $[(\cdot)(1,1,1), (\cdot)(2,2,2), \ldots, (\cdot)(r,r,r)]^T$.

In (14), Khatri-Rao decomposition converts KR-MLR into tensor space, where $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are applied to three different

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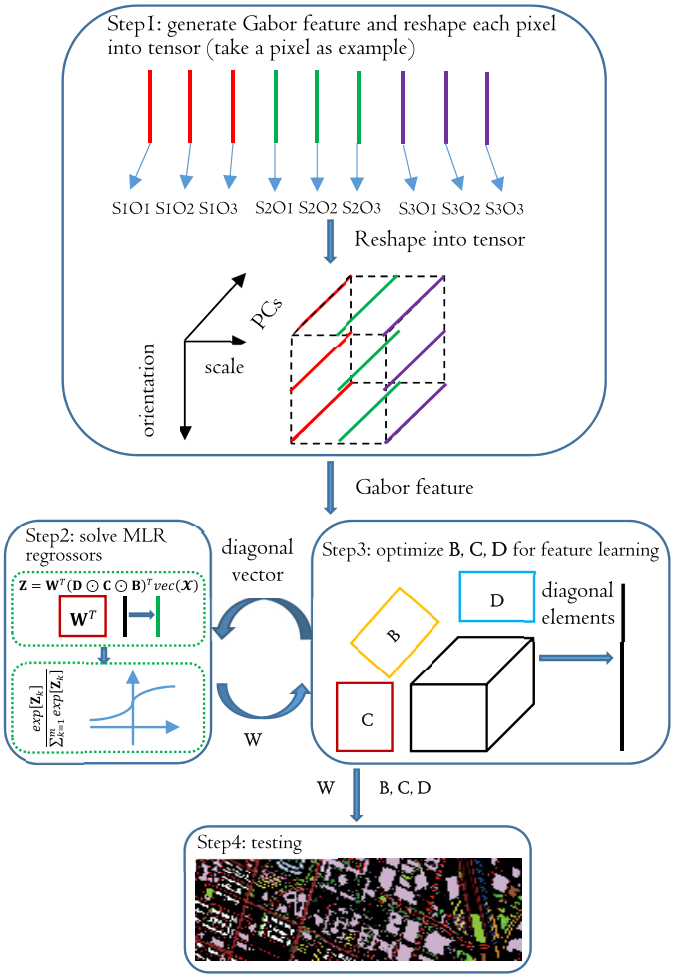


Fig. 1. Workflow of Gabor feature generation and KR-MLR.

axes individually. The number of unknown elements decreases from $d_1d_2d_3r$ to $(d_1+d_2+d_3)r$, and thus, the computation cost is significantly reduced. Take $\mathbf{B}$ for an example. It involves only a d_1r-by-d_1r (about 1000-by-1000) matrix inversion. It is feasible in practice.

Fig. 1 shows the workflow of KR-MLR. Step 1 illustrates Gabor feature generation with three scales and three orientations and its tensor transform. Steps 2 and 3 describe the key to KR-MLR. Step 2 is classifier training, which updates the regressor $\mathbf{w}$. Step 3 is the proposed Khatri-Rao decomposition-based feature learning, which outputs a vector as an input of step 2. Both will be detailed in Section III-C to III-F. Step 4 is the test process.

Furthermore, the class information should be taken into consideration to solve the second disadvantage of Gabor feature. It is expected that the samples from the same class are as close as possible after feature learning. Thus, a discriminant regularization is employed as prior together with Gaussian prior for $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$

$$p(\mathbf{w}) \propto \exp\Bigg(-\frac{\lambda}{2}\|\mathbf{w}\|_2^2 - \frac{\beta_1}{4}\sum_{j=1}^{n}\sum_{i=1}^{n}\gamma_{ij} \times \Big\|\mathrm{diag}\big(\mathfrak{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T\big) - \mathrm{diag}\big(\mathfrak{X}_j \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T\big)\Big\|_2^2 - \frac{\beta_2}{2}\big(\|\mathbf{B}\|_F^2 + \|\mathbf{C}\|_F^2 + \|\mathbf{D}\|_F^2\big)\Bigg) \tag{15}$$

where β_1 and β_2 are used as balance factors in (15). γ_{ij}, entry of matrix $\mathbf{\Gamma}$, is the similarity between $\mathfrak{X}_i$ and $\mathfrak{X}_j$. A typical definition of γ_{ij} is

$$\gamma_{ij} = \begin{cases} \dfrac{\langle \mathfrak{X}_i, \mathfrak{X}_j \rangle}{||\mathfrak{X}_i||_F ||\mathfrak{X}_j||_F}, & y_i = y_j \text{ and } \mathfrak{X}_i, \mathfrak{X}_j \text{ are neighbors} \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

where $\langle \mathfrak{X}_i, \mathfrak{X}_j \rangle = (\mathrm{vec}(\mathfrak{X}_i))^T \mathrm{vec}(\mathfrak{X}_j)$ is the tensor inner product. $\mathrm{vec}(\cdot)$ means vectorization. $\|\mathfrak{X}_i\|_F = (\langle \mathfrak{X}_i, \mathfrak{X}_i \rangle)^{1/2}$.

KR-MLR brings about several remarkable benefits. First, it builds a joint optimization mechanism for both feature learning and classifier training. It is prone to get more effective spatial feature in classification sense since classification is the optimization objective. Second, tensor analysis and Khatri-Rao decomposition techniques greatly reduce the computation cost and ensure the availability of KR-MLR. Third, Laplacian regularization enhances the discriminant ability. Consequently, KR-MLR boosts classification performance at a relatively low cost.

C. KR-MLR Optimization: Update of $\mathbf{w}$

KR-MLR is optimized in iterative fashion. That is to optimize each one in $\mathbf{w}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ successively with the other fixed. Let $\mathbf{I}_{mr}$ be a mr-by-mr identity matrix. Let $\mathbf{1}_m$ be a m-vector of 1s. For $\mathbf{w}$, its solution has been discussed in literature as below [13], [45]

$$\mathbf{w}_{t+1} = (\mathbf{H} - \lambda \mathbf{I}_{mr})^{-1}(\mathbf{H}\mathbf{w}_t - g(\mathbf{w}_t)) \tag{17}$$

where the gradient is given by $g(\mathbf{w}_t) = \sum_{i=1}^{n}(\mathbf{e}_{y_i} - \mathbf{p}_i) \otimes \mathrm{diag}(\mathfrak{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T)$. $\mathbf{e}_{y_i}$ is the y_ith column of $\mathbf{I}_m$ (m-by-m identity matrix). $\mathbf{p}_i$ is the probability output of $\mathfrak{X}_i$ at $\mathbf{w}_t$, and the Hessian matrix $\mathbf{H}$ is a little complicated, which is approximated by

$$\mathbf{H} = -\frac{1}{2}\left[\mathbf{I}_m - \frac{\mathbf{1}_m\mathbf{1}_m^T}{m+1}\right] \otimes \sum_{i=1}^{n} \mathrm{diag}\big(\mathfrak{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T\big) \cdot \big(\mathrm{diag}\big(\mathfrak{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T\big)\big)^T. \tag{18}$$

D. Update of $\mathbf{B}$

In the following, the update process of the introduced unknown matrices $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ will be detailed. Let $l(\mathbf{B})$ denote the log-likelihood function of the proposed KR-MLR with respect to $\mathbf{B}$

$$l(\mathbf{B}) = \sum_{i=1}^{n}\Bigg(\mathbf{w}^{(y_i)^T}\mathrm{diag}\big(\mathfrak{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T\big) - \log\sum_{k=1}^{m}\Big[\mathbf{w}^{(k)^T}\mathrm{diag}\big(\mathfrak{X}_i \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \times_3 \mathbf{D}^T\big)\Big]\Bigg). \tag{19}$$

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For the convenience of derivation, (19) is reshaped by 1-mode flattening of $\mathcal{X}_i$

$$l(\mathbf{B}) = \sum_{i=1}^{n}\left(\mathbf{w}^{(y_i)^T}\operatorname{diag}\left(\mathbf{B}^T\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C})\right) - \log\sum_{k=1}^{m}\left[\mathbf{w}^{(k)^T}\operatorname{diag}\left(\mathbf{B}^T\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C})\right)\right]\right). \tag{20}$$

Its gradient can be found by seeking the first-order derivative as

$$\nabla l(\mathbf{B}) = \sum_{i=1}^{n}\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C})\left(\operatorname{diag}\left(\mathbf{w}^{(y_i)}\right) - \sum_{k=1}^{m}p_{ik}\operatorname{diag}\left(\mathbf{w}^{(k)}\right)\right) \tag{21}$$

where p_{ik} is the kth element of $\mathbf{p}_i$. For simplicity, its vectorization version is deduced as

$$\operatorname{vec}(\nabla l(\mathbf{B})) = \operatorname{sum}\left(\mathbf{V}_B\circ\left[\left(\mathbf{W}\otimes\mathbf{1}_{d_1}\right)(\mathbf{Y}-\mathbf{P})\right],2\right) \tag{22}$$

where $\mathbf{V}_B$ is the auxiliary variable and defined as $\mathbf{V}_B = [\operatorname{vec}(\mathbf{X}_{1(1)}(\mathbf{D}\odot\mathbf{C})), \operatorname{vec}(\mathbf{X}_{2(1)}(\mathbf{D}\odot\mathbf{C})), \ldots, \operatorname{vec}(\mathbf{X}_{n(1)}\cdot(\mathbf{D}\odot\mathbf{C}))]$. $\circ$ is the Hadamard product, which is defined as elementwise multiplication of two matrices of the same size. $\operatorname{sum}(\cdot,2)$ is the sum operator with respect to rows of the given matrix. $\mathbf{W}, \mathbf{Y}$, and $\mathbf{P}$ are individually defined as $\mathbf{W} = [\mathbf{w}^{(1)}, \mathbf{w}^{(2)}, \ldots, \mathbf{w}^{(m)}]$, $\mathbf{Y} = [\mathbf{e}_{y_1}, \mathbf{e}_{y_2}, \ldots, \mathbf{e}_{y_n}]$, and $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_n]$. $\mathbf{1}_{d_1}$ is d_1-vector of 1s.

In addition, its Hessian matrix can be given by seeking the second-order derivative as

$$\nabla^2 l(\mathbf{B}) = -\sum_{i=1}^{n}\left(\left(\mathbf{W}\otimes\mathbf{1}_{d_1}\right)\left[\operatorname{diag}(\mathbf{p}_i)-\mathbf{p}_i\mathbf{p}_i^T\right]\left(\mathbf{W}\otimes\mathbf{1}_{d_1}\right)^T\right) \circ\left[\operatorname{vec}\left(\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C})\right)\left(\operatorname{vec}\left(\mathbf{X}_{i(1)}(\mathbf{D}\odot\mathbf{C})\right)\right)^T\right]. \tag{23}$$

In (23), there exists n Hadamard product and summation as well as one potential matrix inversion [in (27)] during each iteration. It is time-consuming in computation. To reduce the computation complexity, this article approximates it using some trick in [45]

Lemma 1 ([45]): $\operatorname{diag}(\mathbf{p}_i)-\mathbf{p}_i\mathbf{p}_i^T \preccurlyeq [\mathbf{I}_m-\mathbf{1}_m\mathbf{1}_m^T/(m+1)]/2$, where $\preccurlyeq$ is the symbol of semidefinite.

By above lemma, $\nabla^2 l(\mathbf{B})$ can be approximated by

$$\nabla^2 l(\mathbf{B}) = -\left(\left(\mathbf{W}\otimes\mathbf{1}_{d_1}\right)\left[\mathbf{I}_m-\mathbf{1}_m\mathbf{1}_m^T/(m+1)\right]/2\left(\mathbf{W}\otimes\mathbf{1}_{d_1}\right)^T\right) \circ\left(\mathbf{V}_B\mathbf{V}_B^T\right). \tag{24}$$

Next, a simple proof is given to illustrate that (24) is the lower bound of (23), i.e., $\mathbf{G}\circ\mathbf{F}\succcurlyeq\mathbf{E}\circ\mathbf{F}$ if $\mathbf{G}\succcurlyeq\mathbf{E}$ and $\mathbf{F}$ is a semidefinite matrix. $\succcurlyeq$ is the symbol of semidefinite. Let chol be Cholesky decomposition and $\mathbf{x}$ be any column vector, then

$$\begin{aligned}
&\mathbf{x}^T[(\mathbf{G}-\mathbf{E})\circ\mathbf{F}]\mathbf{x}\\
&= \operatorname{tr}\left[\operatorname{diag}(\mathbf{x})(\mathbf{G}-\mathbf{E})\operatorname{diag}(\mathbf{x})\mathbf{F}\right]\\
&= \operatorname{tr}\left[\operatorname{diag}(\mathbf{x})\operatorname{chol}(\mathbf{G}-\mathbf{E})(\operatorname{chol}(\mathbf{G}-\mathbf{E}))^T\right.\\
&\qquad\left.\times\operatorname{diag}(\mathbf{x})\operatorname{chol}(\mathbf{F})(\operatorname{chol}(\mathbf{F}))^T\right]\\
&= \operatorname{tr}\left[(\operatorname{chol}(\mathbf{F}))^T\operatorname{diag}(\mathbf{x})\operatorname{chol}(\mathbf{G}-\mathbf{E})\right.\\
&\qquad\left.\times\left((\operatorname{chol}(\mathbf{F}))^T\operatorname{diag}(\mathbf{x})\operatorname{chol}(\mathbf{G}-\mathbf{E})\right)^T\right]\geq 0.
\end{aligned}$$ ■

Next, the Laplacian regularization can be simplified as its matrix form

$$\begin{aligned}
&\sum_{j=1}^{n}\sum_{i=1}^{n}\gamma_{ij}\left\|\operatorname{diag}\left(\mathcal{X}_i\times_1\mathbf{B}^T\times_2\mathbf{C}^T\times_3\mathbf{D}^T\right)\right.\\
&\qquad\left.-\operatorname{diag}\left(\mathcal{X}_j\times_1\mathbf{B}^T\times_2\mathbf{C}^T\times_3\mathbf{D}^T\right)\right\|_2^2\\
&= 2(\operatorname{vec}(\mathbf{B}))^T\left[\left(\mathbf{V}_B\mathbf{\Delta}\mathbf{V}_B^T\right)\circ\left(\mathbf{I}_r\otimes\left(\mathbf{1}_{d_1}\mathbf{1}_{d_1}^T\right)\right)\right]\operatorname{vec}(\mathbf{B})
\end{aligned} \tag{25}$$

where Laplacian matrix is defined by $\mathbf{\Delta} = \operatorname{diag}(\sum_{j=1}^{n}\gamma_{ij})-\mathbf{\Gamma}$, and denote $\mathbf{L}_B = (\mathbf{V}_B\mathbf{\Delta}\mathbf{V}_B^T)\circ(\mathbf{I}_r\otimes(\mathbf{1}_{d_1}\mathbf{1}_{d_1}^T))$.

Then, the lower bound of objective function with respective to $\mathbf{B}$ can be written by its second-order Taylor expression at $\mathbf{B}_t$ for $t = 1, 2, \ldots$ (ignore constant term)

$$\begin{aligned}
Q(\mathbf{B}) &= (\operatorname{vec}(\mathbf{B})-\operatorname{vec}(\mathbf{B}_t))^T\operatorname{vec}(\nabla l(\mathbf{B}_t))\\
&\quad+\frac{1}{2}(\operatorname{vec}(\mathbf{B})-\operatorname{vec}(\mathbf{B}_t))^T\nabla^2 l(\mathbf{B}_t)(\operatorname{vec}(\mathbf{B})-\operatorname{vec}(\mathbf{B}_t))\\
&\quad-\frac{\beta_1}{2}(\operatorname{vec}(\mathbf{B}))^T\mathbf{L}_B\operatorname{vec}(\mathbf{B})-\frac{\beta_2}{2}\|\operatorname{vec}(\mathbf{B})\|_2^2.
\end{aligned} \tag{26}$$

The maximizer of (26) is given by

$$\begin{aligned}
\operatorname{vec}(\mathbf{B}_{t+1}) &= \left[\nabla^2 l(\mathbf{B}_t)-\beta_1\mathbf{L}_B-\beta_2\mathbf{I}_{d_1 r}\right]^{-1}\\
&\quad\times\left[\nabla^2 l(\mathbf{B}_t)\operatorname{vec}(\mathbf{B}_t)-\operatorname{vec}(\nabla l(\mathbf{B}_t))\right].
\end{aligned} \tag{27}$$

E. Update of $\mathbf{C}$

Because of symmetry, $\mathbf{C}$ can be updated by the same way. The only difference lies that the log-likelihood function of the proposed KR-MLR with respect to $\mathbf{C}$ is written by 2-mode flattening of $\mathcal{X}_i$

$$l(\mathbf{C}) = \sum_{i=1}^{n}\left(\mathbf{w}^{(y_i)^T}\operatorname{diag}\left(\mathbf{C}^T\mathbf{X}_{i(2)}(\mathbf{D}\odot\mathbf{B})\right) - \log\sum_{k=1}^{m}\left[\mathbf{w}^{(k)^T}\operatorname{diag}\left(\mathbf{C}^T\mathbf{X}_{i(2)}(\mathbf{D}\odot\mathbf{B})\right)\right]\right). \tag{28}$$

Thus, its gradient and Hessian matrix can be found by seeking the first- and second-order derivative by the following equations:

$$\operatorname{vec}(\nabla l(\mathbf{C})) = \operatorname{sum}\left(\mathbf{V}_C\circ\left[\left(\mathbf{W}\otimes\mathbf{1}_{d_2}\right)(\mathbf{Y}-\mathbf{P})\right],2\right) \tag{29}$$

$$\nabla^2 l(\mathbf{C}) = -\left(\left(\mathbf{W}\otimes\mathbf{1}_{d_2}\right)\left[\mathbf{I}_m-\mathbf{1}_m\mathbf{1}_m^T/(m+1)\right]/2\left(\mathbf{W}\otimes\mathbf{1}_{d_2}\right)^T\right)\circ\left(\mathbf{V}_C\mathbf{V}_C^T\right) \tag{30}$$

where $\mathbf{V}_C = [\operatorname{vec}(\mathbf{X}_{1(2)}(\mathbf{D}\odot\mathbf{B})), \operatorname{vec}(\mathbf{X}_{2(2)}(\mathbf{D}\odot\mathbf{B})), \ldots, \operatorname{vec}(\mathbf{X}_{n(2)}(\mathbf{D}\odot\mathbf{B}))]$ is the auxiliary variable.

Then, the lower bound of objective function with respective to $\mathbf{C}$ can be written as

$$\begin{aligned}
Q(\mathbf{C}) &= (\operatorname{vec}(\mathbf{C})-\operatorname{vec}(\mathbf{C}_t))^T\operatorname{vec}(\nabla l(\mathbf{C}_t))\\
&\quad+\frac{1}{2}(\operatorname{vec}(\mathbf{C})-\operatorname{vec}(\mathbf{C}_t))^T\nabla^2 l(\mathbf{C}_t)(\operatorname{vec}(\mathbf{C})-\operatorname{vec}(\mathbf{C}_t))\\
&\quad-\frac{\beta_1}{2}(\operatorname{vec}(\mathbf{C}))^T\mathbf{L}_C\operatorname{vec}(\mathbf{C})-\frac{\beta_2}{2}\|\operatorname{vec}(\mathbf{C})\|_2^2
\end{aligned} \tag{31}$$

where the temp variable $\mathbf{L}_C$ is defined by $\mathbf{L}_C = (\mathbf{V}_C\mathbf{\Delta}\mathbf{V}_C^T)\circ(\mathbf{I}_r\otimes(\mathbf{1}_{d_2}\mathbf{1}_{d_2}^T))$.

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The maximizer of (31) is given by

$$\operatorname{vec}(\mathbf{C}_{t+1}) = \left[\nabla^2 l(\mathbf{C}_t) - \beta_1 \mathbf{L}_C - \beta_2 \mathbf{I}_{d_2 r}\right]^{-1} \times \left[\nabla^2 l(\mathbf{C}_t)\operatorname{vec}(\mathbf{C}_t) - \operatorname{vec}(\nabla l(\mathbf{C}_t))\right]. \quad (32)$$

F. Update of D

Similarly, $\mathbf{D}$ can be updated by the same way. By 3-mode flattening of $\mathfrak{X}_i$, the log-likelihood function of the proposed KR-MLR with respect to $\mathbf{D}$ can be written by

$$l(\mathbf{D}) = \sum_{i=1}^{n}\left(\mathbf{w}^{(y_i)^T}\operatorname{diag}\left(\mathbf{D}^T\mathbf{X}_{i(3)}(\mathbf{C}\odot\mathbf{B})\right) - \log\sum_{k=1}^{m}\left[\mathbf{w}^{(k)^T}\operatorname{diag}\left(\mathbf{D}^T\mathbf{X}_{i(3)}(\mathbf{C}\odot\mathbf{B})\right)\right]\right). \quad (33)$$

Thus, its gradient and Hessian matrix can be found by seeking the first- and second-order derivative by the following equations:

$$\operatorname{vec}(\nabla l(\mathbf{D})) = \operatorname{sum}\left(\mathbf{V}_D \circ \left[\left(\mathbf{W}\otimes\mathbf{1}_{d_3}\right)(\mathbf{Y}-\mathbf{P})\right], 2\right) \quad (34)$$

$$\nabla^2 l(\mathbf{D}) = -\left(\left(\mathbf{W}\otimes\mathbf{1}_{d_3}\right)\left[\mathbf{I}_m - \mathbf{1}_m\mathbf{1}_m^T/(m+1)\right]/2\left(\mathbf{W}\otimes\mathbf{1}_{d_3}\right)^T\right)\circ\left(\mathbf{V}_D\mathbf{V}_D^T\right) \quad (35)$$

where $\mathbf{V}_D = [\operatorname{vec}(\mathbf{X}_{1(3)}(\mathbf{C}\odot\mathbf{B})), \operatorname{vec}(\mathbf{X}_{2(3)}(\mathbf{C}\odot\mathbf{B})), \ldots, \operatorname{vec}(\mathbf{X}_{n(3)}(\mathbf{C}\odot\mathbf{B}))]$ is the auxiliary variable.

Then, the lower bound of objective function with respective to $\mathbf{D}$ can be written as

$$Q(\mathbf{D}) = (\operatorname{vec}(\mathbf{D}) - \operatorname{vec}(\mathbf{D}_t))^T\operatorname{vec}(\nabla l(\mathbf{D}_t)) + \frac{1}{2}(\operatorname{vec}(\mathbf{D})-\operatorname{vec}(\mathbf{D}_t))^T\nabla^2 l(\mathbf{D}_t)(\operatorname{vec}(\mathbf{D})-\operatorname{vec}(\mathbf{D}_t)) - \frac{\beta_1}{2}(\operatorname{vec}(\mathbf{D}))^T\mathbf{L}_D\operatorname{vec}(\mathbf{D}) - \frac{\beta_2}{2}\|\operatorname{vec}(\mathbf{D})\|_2^2 \quad (36)$$

where the temp variable $\mathbf{L}_D$ is defined by $\mathbf{L}_D = (\mathbf{V}_D\boldsymbol{\Delta}\mathbf{V}_D^T)\circ(\mathbf{I}_r\otimes(\mathbf{1}_{d_3}\mathbf{1}_{d_3}^T))$.

The maximizer of (36) is given by

$$\operatorname{vec}(\mathbf{D}_{t+1}) = \left[\nabla^2 l(\mathbf{D}_t) - \beta_1\mathbf{L}_D - \beta_2\mathbf{I}_{d_3 r}\right]^{-1} \times \left[\nabla^2 l(\mathbf{D}_t)\operatorname{vec}(\mathbf{D}_t) - \operatorname{vec}(\nabla l(\mathbf{D}_t))\right]. \quad (37)$$

The whole solving procedure is summarized in Algorithm 1.

Algorithm 1 KR-MLR Optimization Procedure

Input: training data $\{\mathfrak{X}_i\}_{i=1}^n$, parameters λ, β_1, β_2

1. Initialize matrices $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ with random values.
2. While not converged do
   - 2.1) while $t \le iter1$ do
     - update regressors $\mathbf{w}_{t+1}$ by equation (17)
     - end while
   - 2.2) while $t \le iter2$ do
     - update gradient $\operatorname{vec}(\nabla l(\mathbf{B}_t))$ by equation (22)
     - update Hessian matrix $\nabla^2 l(\mathbf{B}_t)$ by equation (24)
     - update matrix $\mathbf{B}_{t+1}$ by equation (27)
     - end while
   - 2.3) while $t \le iter2$ do
     - update gradient $\operatorname{vec}(\nabla l(\mathbf{C}_t))$ by equation (29)
     - update Hessian matrix $\nabla^2 l(\mathbf{C}_t)$ by equation (30)
     - update matrix $\mathbf{C}_{t+1}$ by equation (32)
     - end while
   - 2.4) while $t \le iter2$ do
     - update gradient $\operatorname{vec}(\nabla l(\mathbf{D}_t))$ by equation (34)
     - update Hessian matrix $\nabla^2 l(\mathbf{D}_t)$ by equation (35)
     - update matrix $\mathbf{D}_{t+1}$ by equation (37)
     - end while

End while

IV. Experimental Results

Below, an empirical study is developed to illustrate the effectiveness of KR-MLR for HSI classification. It involves four popular HSI datasets, Houston 2018 image, Xiong'an image, Houston 2013 image, and Indian Pines image. For each of them, the following algorithms are utilized for comparison.

1) MLR and support vector machine (SVM) [46]. Both use the same Gabor feature as an input.
2) IAP [38]. It is the latest spatial feature extraction technique.
3) MLRsub-MLL. It is an extended version of MLRsub by graph cut [14], [47].
4) Support tensor machine (STM) [39] and generalized tensor regression (GTR) [40]. Both are tensor-based algorithms.
5) SSRN [16] and fast-3D-CNN [17]. Both are the latest deep learning algorithms.

A. Houston 2018

The first HSI dataset employed is Houston 2018 image, which was acquired by IRTES CASI-1500 sensor over the University of Houston and its neighborhood in Houston, TX, USA, in February of 2017 [48]. The image is across a wide spectrum wavelength range from 380 to 1050 nm, divided into 50 bands. Its spatial dimension is 601 × 2384, where 504 712 pixels are labeled ones. The ground-truth contains 20 different land-cover classes. Here, 1595 labeled pixels, 80 per class, are randomly chosen for training with the remaining for testing. For Unpaved Parking Lots, half labeled pixels are used for training. Regarding to Gabor feature, it is generated with six scales and eight orientations based on the first 48 PCs.

The classification results produced by KR-MLR and compared algorithms are reported in Table I via class specific accuracy, overall accuracy (OA), average accuracy (AA), and κ coefficient. All results are averaged by ten random runs, and the visual maps corresponding to one of them are shown in Fig. 2. Regarding to parameters setting for KR-MLR, the width r of matrices $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ is fixed to 40, while the optimal values for λ, β_1, and β_2 are fixed to $\lambda = 10^5$, $\beta_1 = 10^{-7}$, and $\beta_2 = 10^{-5}$, respectively.

In Table I, the proposed KR-MLR yields the best classification accuracy among all compared algorithms. For example, in terms of OA, it gets considerable improvements than MLR as high as 13.14%. It outperforms IAP, fast-3D-CNN, and

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TABLE I
CLASSIFICATION ACCURACY (%) FOR THE HOUSTON 2018 IMAGE

Class	MLR	MLRsub-MLL	SVM	IAP	STM	GTR	fast-3D-CNN	SSRN	KR-MLR
1	76.76	78.90	82.51	89.91	86.34	35.07	86.00	72.76	86.48
2	71.77	65.16	75.46	85.85	82.37	80.81	73.67	93.36	77.13
3	99.96	99.47	100.00	99.66	99.88	94.30	99.63	86.22	100.00
4	88.28	82.57	85.41	94.41	94.24	45.00	92.53	83.47	95.16
5	86.80	34.59	80.19	89.01	90.87	21.27	86.39	56.74	90.38
6	98.19	92.77	97.59	99.70	98.23	30.27	96.93	79.72	99.99
7	99.03	99.56	100.00	100.00	99.62	91.30	99.24	93.87	99.94
8	74.32	27.76	81.57	83.61	80.56	53.04	67.62	64.72	90.27
9	56.92	82.34	54.61	65.92	74.47	96.93	61.13	98.09	77.17
10	39.63	25.72	42.97	40.07	50.45	58.44	30.13	57.59	46.88
11	35.98	45.28	38.56	40.63	47.31	51.15	37.69	53.60	48.54
12	74.44	6.70	72.30	67.03	58.06	3.01	48.65	6.28	80.57
13	62.37	17.10	56.78	57.79	46.97	63.13	43.39	67.17	67.06
14	98.42	66.59	96.61	91.85	90.09	16.80	87.67	53.90	99.01
15	97.74	46.85	95.54	99.18	96.98	42.91	96.38	89.36	99.23
16	91.33	20.27	80.00	90.40	86.84	56.56	85.04	80.22	94.77
17	100.00	92.97	100.00	100.00	100.00	0.40	99.86	55.65	100.00
18	94.12	55.58	84.12	93.01	87.17	31.16	89.03	44.10	95.40
19	97.55	16.24	93.92	95.98	94.17	18.45	87.10	67.36	96.08
20	96.07	99.98	95.24	97.75	98.47	25.50	96.21	65.39	99.89
OA	62.35	59.66	61.55	68.21	71.63	50.34	60.99	73.69	75.49
AA	81.98	57.82	80.67	84.09	83.15	45.78	78.22	68.48	87.20
κ	0.554	0.477	0.546	0.618	0.651	0.429	0.536	0.679	0.697

even SSRN as well. It should be noted that all of them are widely used benchmarks for HSI classification in recent feature extraction and deep neural network-based works. In Fig. 2, KR-MLR produces a much smoother visual map than others as expected. These results are owed to that KR-MLR provides a joint optimization framework to learn more effective spatial feature along with regressor of classifier, which maximizes their synergy effect. Besides, discriminant Laplacian regularization further enhances the classification performance.

The next experiment is to demonstrate how Gabor feature-related parameters affect the classification accuracy of KR-MLR. In this part, the training set and parameters follow the setting that generates Table I. Fig. 3 shows the surface of OA variation versus scale and orientation number. When scale number is greater than 5 and orientation number is greater than 5, KR-MLR generates very high OAs. Thus, in all experiments, the scale and orientation numbers are fixed to 6 and 8, respectively. Fig. 4 shows the OA variation curve versus window size of Gabor filter for Houston 2018 in blue. It reports the highest OA when window size is set to 35. However, a large window size usually leads oversmooth phenomenon. Fig. 5 shows the classification maps of the whole image of KR-MLR when window size is at 35, 25, and 15. When the window size is decreased from 35 to 15, oversmooth phenomenon is reduced. Accordingly, the consistency of the homogeneous regions become poor, and the classification accuracy decreases from 76.72 to 75.49 and 73.24 (%). To balance accuracy and oversmooth, the window size is set to 25. Since the Gabor feature is generated based on the top PCs, the PCs number chosen in study is critical to classification performance. Fig. 6 shows the OA variation curve versus the

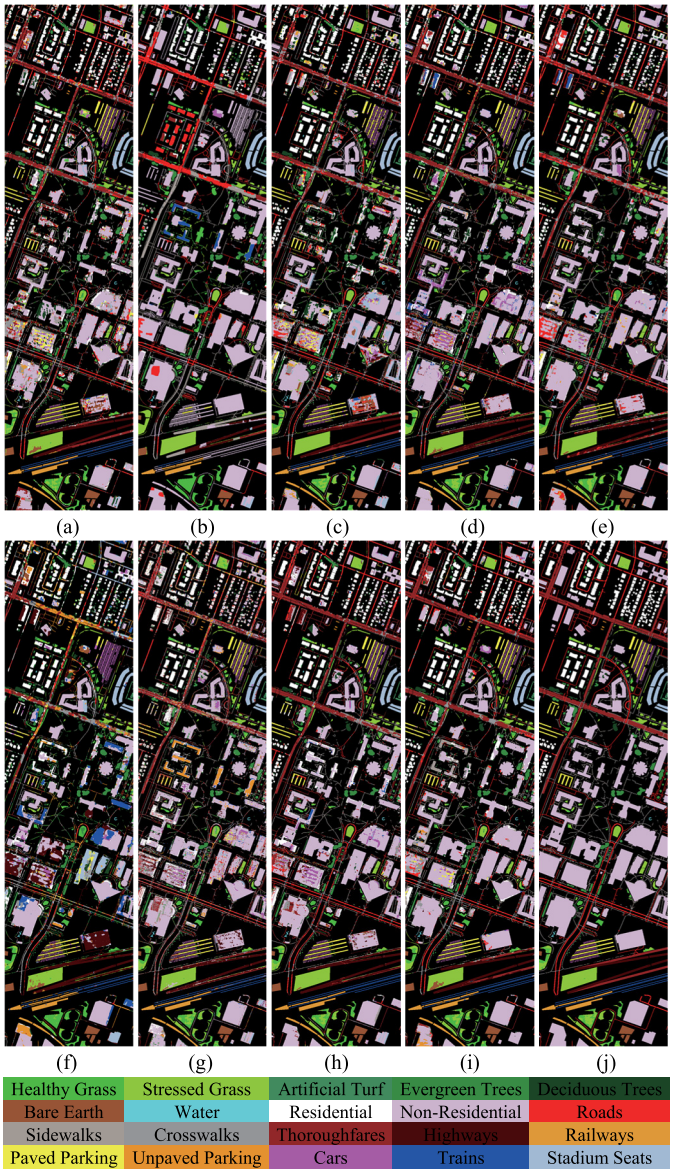


Fig. 2. Visual results for Houston 2018. (a) MLR. (b) MLRsub-MLL. (c) SVM. (d) IAP. (e) STM. (f) GTR. (g) fast-3D-CNN. (h) SSRN. (i) KR-MLR. (j) Ground truth.

number of PCs for Houston 2018 in blue. For this image, it only has 48 nonzero eigenvalues, and Fig. 6 reports the best OA at 48 as well. Thus, all PCs are utilized for Gabor feature generation.

The width r of matrices **B**, **C**, and **D** controls the number of unknown variables and model complexity, playing a significant role in the proposed classifier. Next, an experiment is conducted to investigate its effect on the final performance of KR-MLR. In this part, the training set and parameters follow the setting that generates Table I. A series of seven sets' experiments are performed with r range from 20 to 50. Fig. 7 shows the OA variation curve as a function of r averaged by ten random runs for Houston 2018 in blue line. For this image, the OA curve increases with r at first and then stays stable when $r \geq 40$. Take into consideration the unknown parameters number, $r = 40$ is chosen for Houston 2018 in all related experiments.

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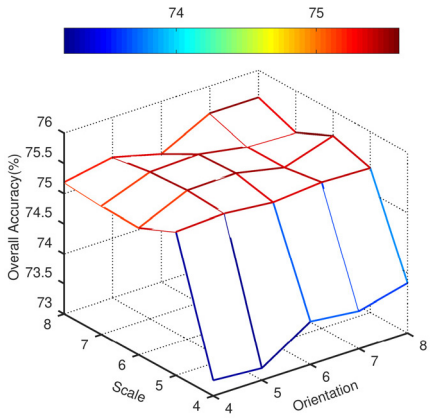


Fig. 3. Effect of scale and orientation number of Gabor filter on KR-MLR for Houston 2018.

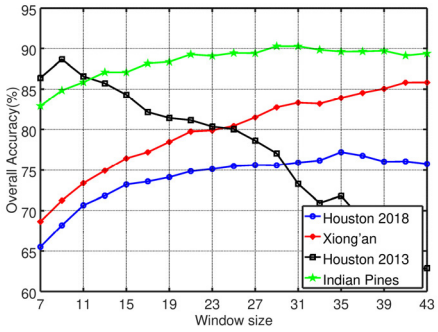


Fig. 4. Effect of window size of Gabor filter on KR-MLR for four images.

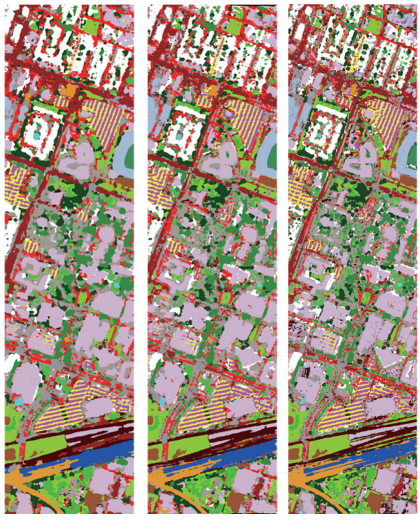


Fig. 5. Classification map of the whole Houston 2018 image by KR-MLR with window size at (a) 35, (b) 25, and (c) 15.

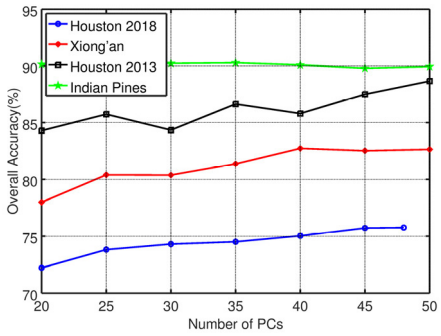


Fig. 6. Effect of PCs number on KR-MLR for four images.

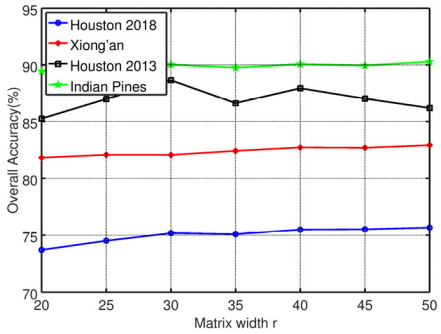


Fig. 7. Effect of matrices width r on KR-MLR for four images.

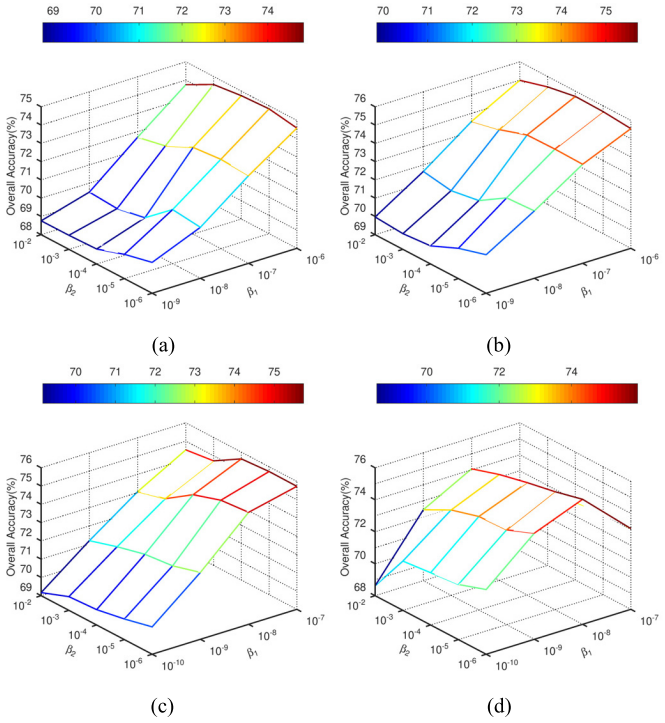


Fig. 8. Effect of β_1 and β_2 on KR-MLR for Houston 2018 with (a) $\lambda = 10^3$, (b) $\lambda = 10^4$, (c) $\lambda = 10^5$, and (d) $\lambda = 10^6$.

As shown in (15), the introduced regularization parameters β_1 and β_2 are of importance for the new classifier KR-MLR. Accordingly, an experiment is performed to study the OA variation with β_1 and β_2. In this part, the same training set is used. It involves four values for λ, i.e., $\lambda = 10^6, 10^5, 10^4, 10^3$. For each λ, the OA variation versus different pairs of β_1 and β_2 is shown in Fig. 8. It is obvious that β_1 leads more remarkable OA variation than β_2, verified by all scenarios in this experiment. By contrast, β_2 provides a fine-tuning means for KR-MLR. Besides, one can find that the maximum of OAs in Fig. 8(a)–(d) appears when $\lambda = 10^5$. Thus, $\lambda = 10^5$ is used for Houston 2018. In Fig. 8(c), it gets the maximum at $\beta_1 = 10^{-7}$ and $\beta_2 = 10^{-5}$.

Finally, it is to study how the training set size affects the classification performance of various methods. Experiments are performed under five different cases with 40–120 labeled

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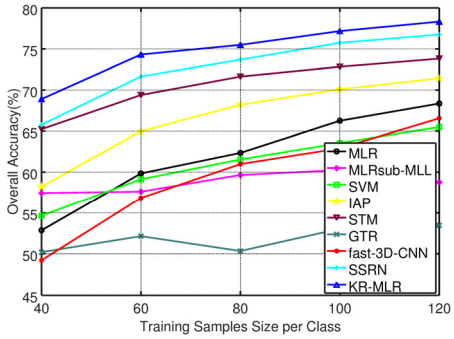


Fig. 9. Effect of training samples size on KR-MLR for Houston 2018.

pixels per class for training and the rest for testing. For unpaved parking lots (even half of the labeled pixels size is still less than required), just half of them are chosen for training. The parameters are set the same as those discussed in previous experiments. Fig. 9 reports the average OAs for all cases by ten random runs, where the x-axis means the training samples size per class while the y-axis means the averaged OAs. From Fig. 9, one can see that the OAs of all methods monotonically increase with the training set size and KR-MLR exceeds all compared ones. Therefore, the proposed KR-MLR model is robust for HSI classification under various conditions.

B. Xiong'an

The second HSI dataset employed is Xiong'an image [49], which was acquired by a visible and near-infrared imaging spectrometer designed by Shanghai Institute of Technical Physics, Chinese Academy of Sciences over Matiwan village in Xiong'an, Hebei province, China, in October 2017. The image was generated with 256 working bands across 400–1000 nm. The image size is 1580 × 3750. It contains 3 341 881 labeled pixels covering 19 ground-truth classes. In this part, 80 labeled samples are chosen from each class individually as training set. The Gabor feature is generated with eight scales and four orientations based on the first 40 PCs.

Table II shows the classification results provided by KR-MLR and all compared ones. All results are averaged by ten independent random runs. The visual maps corresponding to one of them are shown in Fig. 10. For parameters setting for KR-MLR, the width r of matrices **B**, **C**, and **D** is fixed to 40, while the optimal values for λ, β_1, and β_2 are fixed to $\lambda = 10^5$, $\beta_1 = 10^{-8}$, and $\beta_2 = 10^{-4}$, respectively.

In Table II, KR-MLR runs best among all classifiers used for this image. For example, KR-MLR outperforms MLR in OA by 6.6%. It excels STM, IAP, and even SSRN by a large margin as well. In Fig. 10, KR-MLR shows better in neighborhood consistency when compared with others. These results confirm that the joint optimization framework learns more effective spatial feature and classifier concurrently, which maximizes the synergy effect. Besides, discriminant Laplacian regularization further enhances the classification performance.

The parameter analysis about Gabor feature is conducted as follows. In this part, the training set and parameters follow the setting that generates Table II. Fig. 11 shows the surface

TABLE II
CLASSIFICATION ACCURACY (%) FOR THE XIONG'AN IMAGE

Class	MLR	MLRsub-MLL	SVM	IAP	STM	GTR	fast-3D-CNN	SSRN	KR-MLR
1	96.17	98.50	91.77	94.71	97.20	96.03	96.90	98.76	98.30
2	96.61	98.36	92.67	96.19	97.54	68.66	96.39	92.98	98.30
3	94.09	95.57	95.02	95.70	96.38	79.13	95.28	99.57	94.80
4	74.99	62.00	67.42	71.30	84.25	38.36	77.90	56.66	83.51
5	83.12	91.28	65.84	77.57	90.35	93.28	73.56	57.51	89.27
6	97.15	87.09	86.30	90.70	95.45	10.62	92.08	48.41	98.32
7	75.17	56.08	56.77	71.82	79.86	77.96	66.42	87.23	79.05
8	76.09	74.85	62.59	73.37	76.56	74.75	59.60	76.46	83.66
9	92.21	55.85	73.98	83.35	92.18	30.60	84.77	46.95	94.73
10	63.33	71.57	46.56	60.63	71.78	70.62	58.86	81.11	73.67
11	80.41	46.45	79.43	71.35	80.63	26.33	73.43	40.10	88.09
12	68.28	79.99	70.23	66.18	82.44	29.27	61.53	20.44	88.99
13	61.54	53.56	48.11	56.95	70.84	39.02	57.40	44.49	76.39
14	82.95	53.64	65.48	79.20	85.87	29.73	76.76	52.40	88.92
15	64.63	73.46	38.62	53.86	63.38	97.60	54.45	74.54	72.35
16	93.96	32.26	87.06	86.33	90.00	11.24	84.08	39.92	97.38
17	95.61	97.31	83.55	96.09	96.21	15.96	94.40	35.62	97.76
18	98.42	36.47	88.90	85.20	94.11	4.46	87.13	26.64	99.91
19	46.03	98.10	94.19	94.66	93.68	57.49	90.43	50.51	88.96
OA	76.10	76.64	61.06	71.31	78.86	60.26	69.83	71.38	82.74
AA	81.09	71.71	73.39	79.22	86.25	50.06	77.97	59.49	89.07
κ	0.730	0.734	0.570	0.680	0.762	0.564	0.664	0.670	0.804

of OA variation versus scale and orientation number. When scale number is greater than 6 and orientation number is set to 4, KR-MLR generates very satisfactory OAs. Thus, in all experiments, the scale and orientation numbers are fixed to 8 and 4, respectively. The OA variation versus window size and PCs number are reported in Figs. 4 and 6 for Xiong'an in red. It reports the highest OA when window size is set to 41 and PCs number is set to 40. However, with the balance of accuracy and oversmooth in mind, the window size is set to 29, and the PCs number is fixed to 40 for Gabor feature generation.

Besides, the red line in Fig. 7 shows OA variation of KR-MLR with matrices width r of **B**, **C**, and **D** for Xiong'an image under the same experiment setting. In this part, seven sets' experiments are performed to investigate the effect of r with range from 20 to 50. The OA curve is relatively stable and arrives its peak at $r = 40$. Thus, $r = 40$ is directly used for this image in all related experiments.

Next, an experiment is designed to analyze how the introduced regularization parameters β_1 and β_2 affect the final classification performance of KR-MLR. In this part, the same training set as that generates Table II is utilized. It involves four values for λ, i.e., $\lambda = 10^6, 10^5, 10^4, 10^3$. For each λ, the OA variation versus different pairs of (β_1, β_2) is given in Fig. 12. One can see from Fig. 12 that the OA curved surfaces sharply change along with β_1-axis while slightly change along with β_2-axis. By comparison, one can figure out that the maximums of OAs in Fig. 12(a)–(d) show only subtle difference and Fig. 12(c) produces the highest OA. Thus, $\lambda = 10^5$ is used for Xiong'an image. The corresponding optimal parameters' pair for (β_1, β_2) is $(10^{-8}, 10^{-4})$.

Finally, it is to study how the training set size affects the classification performance of various methods. Experiments

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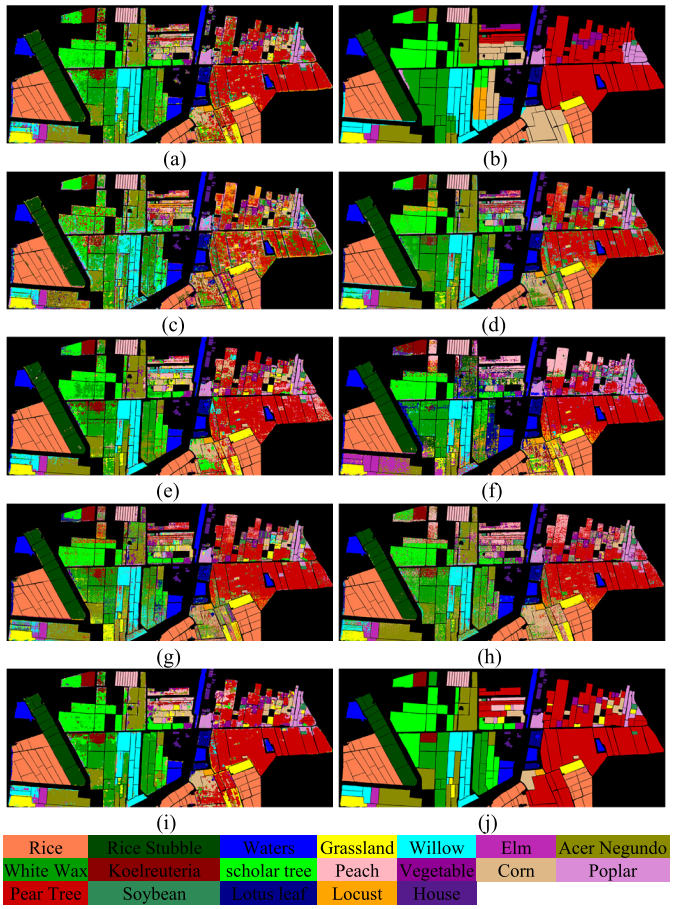


Fig. 10. Visual results for Xiong'an. (a) MLR. (b) MLRsub-MLL. (c) SVM. (d) IAP. (e) STM. (f) GTR. (g) Fast-3D-CNN. (h) SSRN. (i) KR-MLR. (j) Ground truth.

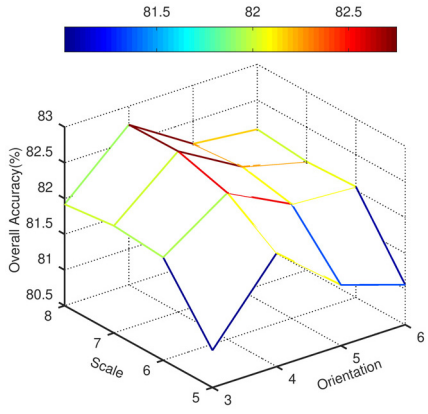


Fig. 11. Effect of scale and orientation number of Gabor filter on KR-MLR for Xiong'an.

are performed under five scenarios with 40–120 labeled pixels chosen from each class as training samples and the rest as test ones. The parameters are set the same as detailed in previous parameter analysis. Fig. 13 reports the average OAs for all cases by ten random runs. From Fig. 13, one can see that KR-MLR is still ahead by a large margin than others, proving its robustness and stability for HSI classification under various conditions.

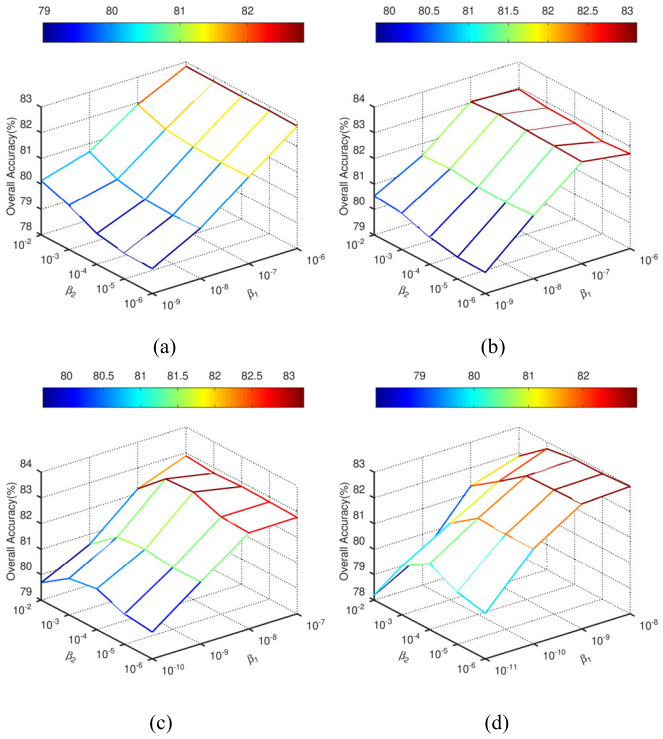


Fig. 12. Effect of β_1 and β_2 on KR-MLR for Xiong'an with (a) $\lambda = 10^3$, (b) $\lambda = 10^4$, (c) $\lambda = 10^5$, and (d) $\lambda = 10^6$.

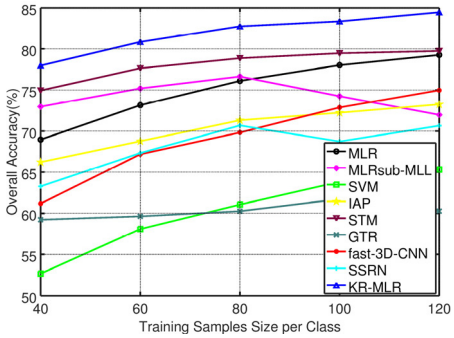


Fig. 13. Effect of training samples size on KR-MLR for Xiong'an.

C. Houston 2013

The third HSI dataset employed is Houston 2013 image, which was acquired by IRTES CASI-1500 sensor over the same area of the University of Houston in Houston, TX, USA, in 2013 [50]. The image is across a wide spectrum wavelength range from 364 to 1046 nm, divided into 144 bands. Its spatial dimension is 349 × 1905, where 15 029 pixels are labeled ones. The ground-truth contains 15 different land-cover classes. Here, 2832 labeled pixels are chosen for training with the remaining for testing. Regarding to Gabor feature, it is generated with four scales and seven orientations based on the first 50 PCs.

Table III reports the classification results of KR-MLR and other methods. Fig. 14 shows the visual maps. For KR-MLR, the parameters are set as follows. The width r of matrices $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ is fixed to 30, while the optimal values for λ, β_1, and β_2 are fixed to $\lambda = 10^6$, $\beta_1 = 10^{-12}$, and $\beta_2 = 10^{-2}$, respectively.

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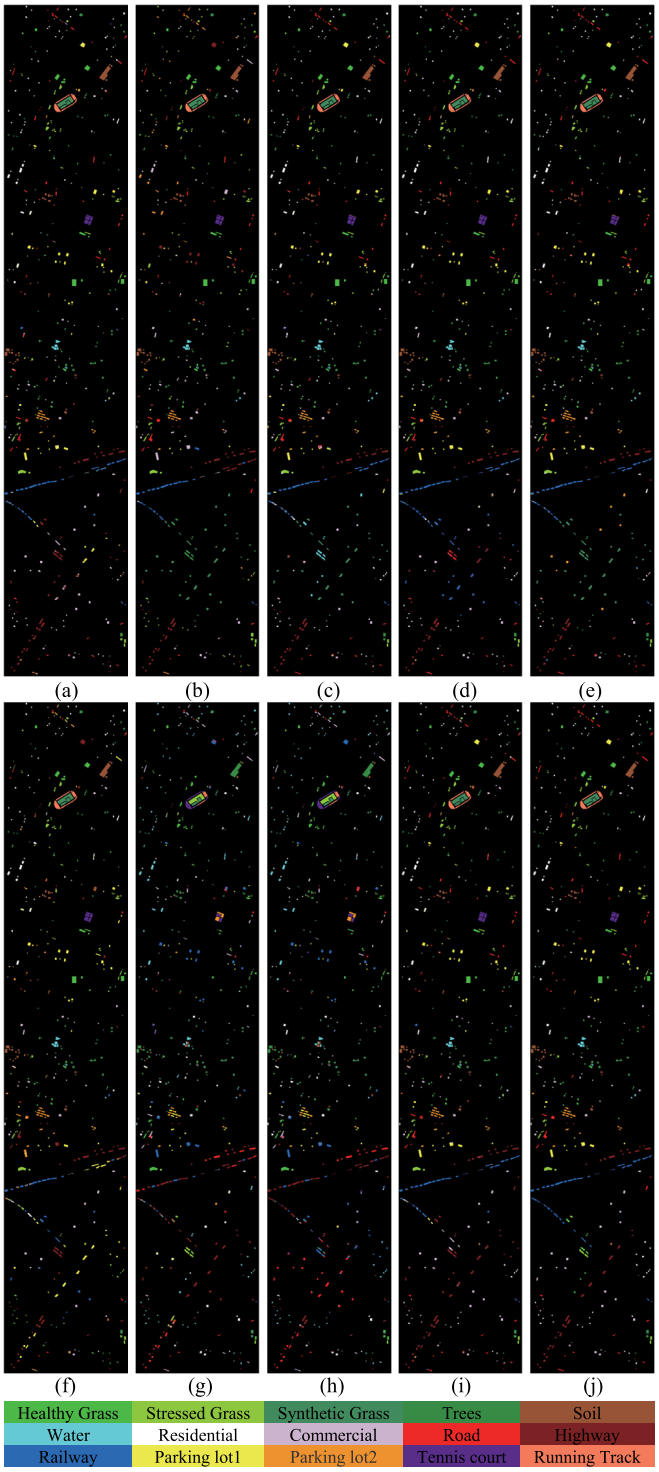


Fig. 14. Visual results for Houston 2013. (a) MLR. (b) MLRsub-MLL. (c) SVM. (d) IAP. (e) STM. (f) GTR. (g) Fast-3D-CNN. (h) SSRN. (i) KR-MLR. (j) Ground truth.

In Table III, IAP is pretty competitive, and KR-MLR yields nearly the same OA. In contrast, MLRsub-MLL produces poor result after adjusting parameters thoroughly. These results verify that the integrated mechanism of KR-MLR boosts regressor and feature learning and thus brings about better classification performance. Besides, discriminant Laplacian regularization provides further improvement.

TABLE III
CLASSIFICATION ACCURACY (%) FOR THE HOUSTON 2013 IMAGE

Class	MLR	MLRsub-MLL	SVM	IAP	STM	GTR	Fast-3D-CNN	SSRN	KR-MLR
1	76.92	80.91	80.91	82.81	81.76	95.81	82.62	99.20	83.09
2	83.92	75.56	81.20	85.06	83.36	74.87	78.94	92.34	85.05
3	93.26	100.00	97.42	99.60	100.00	100.00	95.24	99.80	99.00
4	84.84	66.57	84.46	91.38	78.21	80.47	89.39	93.48	88.35
5	99.52	90.62	98.39	100.00	98.39	84.28	98.67	99.15	99.71
6	79.02	95.80	94.40	98.60	94.40	100.00	93.70	97.85	90.90
7	74.16	1.49	84.23	85.35	83.20	76.63	84.04	93.54	82.64
8	71.98	50.61	60.96	74.45	46.05	87.81	62.01	85.44	83.85
9	74.22	50.42	84.04	83.76	84.04	77.89	68.36	91.16	82.34
10	73.06	68.24	66.79	94.02	64.18	38.17	54.24	55.44	96.33
11	78.55	55.97	70.58	90.32	82.92	61.14	73.81	77.71	79.69
12	92.31	0.00	83.86	90.39	89.91	29.42	92.21	60.09	95.29
13	67.01	54.73	66.31	76.14	81.40	30.66	90.17	83.93	80.35
14	96.35	99.19	94.73	99.60	93.52	100.00	99.59	99.99	100.00
15	97.04	98.94	100.00	99.79	94.92	94.98	99.57	81.94	99.57
OA	82.04	59.08	81.28	88.79	81.22	67.27	80.86	83.70	88.66
AA	82.81	65.94	83.22	90.09	83.75	75.48	84.17	87.40	89.74
κ	0.805	0.559	0.798	0.878	0.797	0.646	0.793	0.823	0.877

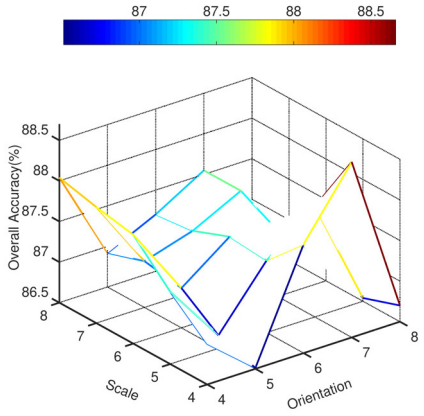


Fig. 15. Effect of scale and orientation number of Gabor filter on KR-MLR for Houston 2013.

The parameter analysis about Gabor feature is conducted as well. In this part, the training set and parameters follow the setting that generates Table III. Fig. 15 shows the OA variation versus scale and orientation number. When scale number is set to 4 and orientation number is greater than 7, KR-MLR generates very competitive OAs. Thus, in all experiments, the scale and orientation numbers are fixed to 4 and 7, respectively. The OA variations versus window size and PCs number are reported in Figs. 4 and 6 for Houston 2013 in black. It shows that OAs decrease when the window size is greater than 9. Thus, 9 is used for experiments. For PCs number, 50 is used since it gets distinct advantage than others.

Next, it is to investigate how the matrices' width r of $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ affects the final classification performance of KR-MLR. The training set and other parameters are set the same as mentioned above. Fig. 7 reports OA as a function of r in black line. For this image, the OA curve fluctuates in the investigated region and gets the maximum at 30. Thus,

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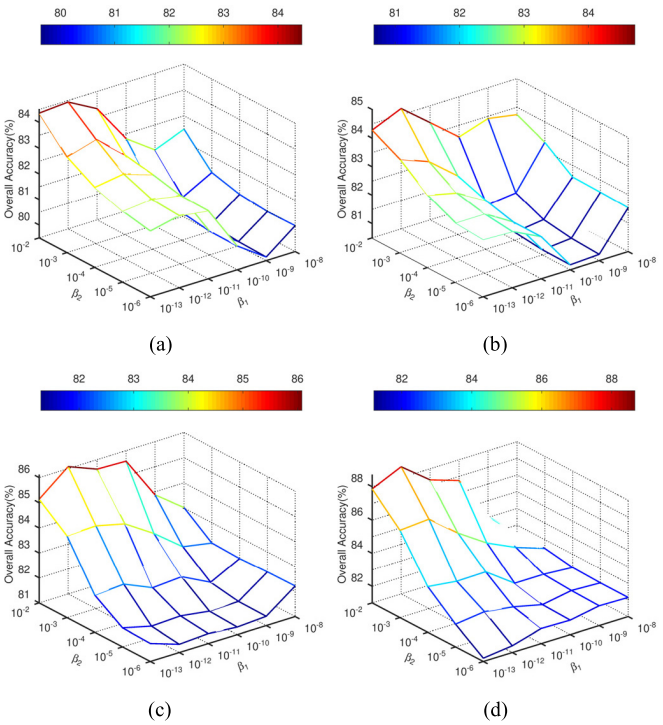


Fig. 16. Effect of β_1 and β_2 on KR-MLR for Houston 2013 with (a) $\lambda = 10^3$, (b) $\lambda = 10^4$, (c) $\lambda = 10^5$, and (d) $\lambda = 10^6$.

all experiments for Houston 2013 directly use 30 for matrices width.

The following experiment is to analyze the OA variation of KR-MLR versus the introduced regularization parameters β_1 and β_2. In this part, the same training set for Table III is employed. Four values are successively set for λ in this study, i.e., $\lambda = 10^6, 10^5, 10^4, 10^3$. Fig. 16 shows the OAs' variation with different pairs of (β_1, β_2) for each λ. From Fig. 16, one can find that β_2 has greater effect on OA than β_1, a little difference from the variation surfaces of Houston 2018 and Xiong'an. It shows that both β_1 and β_2 are critical to KR-MLR. By comparison, the maximums of OAs are obtained at $\lambda = 10^6$. Thus, $\lambda = 10^6$ is used in all experiments. The corresponding optimal parameters' pair for (β_1, β_2) is $(10^{-12}, 10^{-2})$.

D. Indian Pines

The fourth HSI dataset employed is Indian Pines image, which was acquired by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over an agriculture land of Northwestern Indiana in June 1992 [51]. The image is across a wide spectrum wavelength range from 0.4 to 2.5 μm, divided into 220 bands. Among of them, 20 water absorption bands are removed in advance. Its spatial dimension is 145 × 145, where 10 366 pixels are labeled ones. The ground-truth contains 16 different land-cover classes. Here, 303 labeled pixels, 20 per class, are randomly chosen for training with the remaining for testing. For grass-pasture-mowed and oats, half labeled pixels are used for training. Regarding to Gabor feature, it is generated with seven scales and four orientations based on the first 30 PCs.

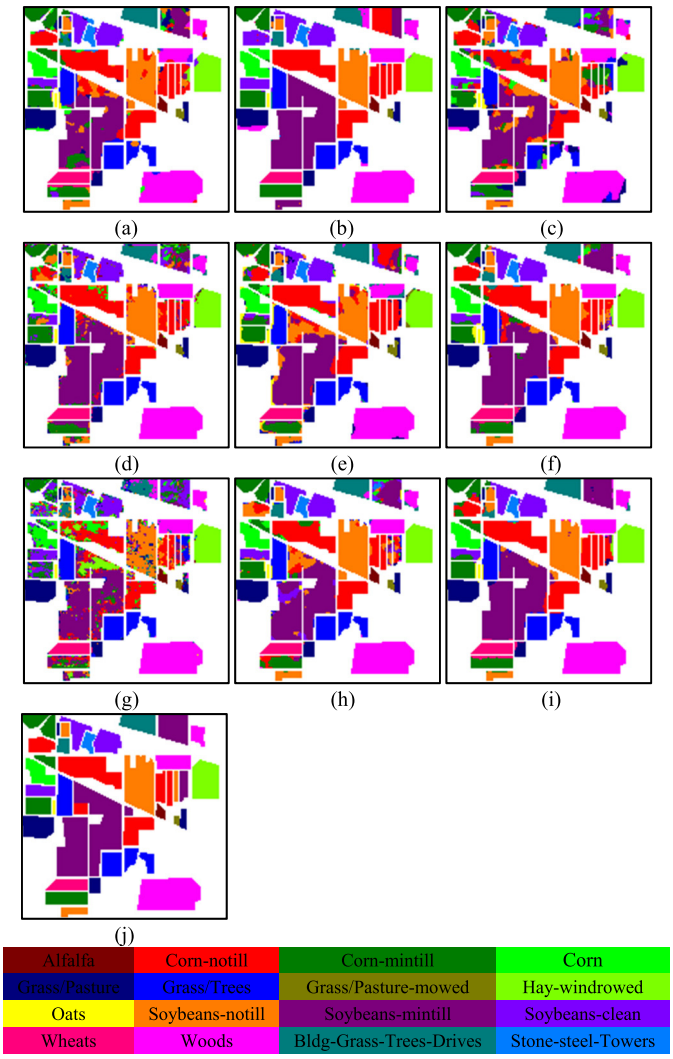


Fig. 17. Visual results for Indian Pines. (a) MLR. (b) MLRsub-MLL. (c) SVM. (d) IAP. (e) STM. (f) GTR. (g) fast-3D-CNN. (h) SSRN. (i) KR-MLR. (j) Ground truth.

The classification results produced by KR-MLR and other algorithms are reported in Table IV. All results are averaged by ten random runs. Besides, the visual maps corresponding to one of them are shown in Fig. 17. For KR-MLR, the width r of matrices **B**, **C**, and **D** is fixed to 40, while the optimal values for λ, β_1, and β_2 are fixed to $\lambda = 10^4$, $\beta_1 = 10^{-7}$, and $\beta_2 = 10^{-8}$, respectively.

In Table IV, KR-MLR yields the best classification accuracy among all compared algorithms. For example, it excels MLR in OA as high as 7.3%. It outperforms STM and GTR as well. In Fig. 17, KR-MLR produces a much smoother visual map than others as expected. These results are owed to that KR-MLR can learn more powerful spatial feature and classifier concurrently, which maximizes the classification performance. Besides, discriminant Laplacian regularization brings about further improvement.

The OA variations versus window size and PCs number are reported in Figs. 4 and 6 for Indian Pines in green. It reports that OAs are very stable for the investigated window size range from 25 to 43 and for PCs number from 20 to 50. Thus, 27 and 30 are used for window size and PCs number, respectively.

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TABLE IV
CLASSIFICATION ACCURACY (%) FOR THE INDIAN PINES IMAGE

Class	MLR	MLRsub-MLL	SVM	IAP	STM	GTR	fast-3D-CNN	SSRN	KR-MLR
1	100.00	96.17	100.00	95.00	92.94	64.96	94.11	85.38	100.00
2	79.17	79.77	64.03	75.91	81.55	89.75	48.68	87.06	83.18
3	86.01	85.46	77.12	67.23	85.76	76.87	45.06	73.58	91.00
4	98.97	99.76	95.88	92.38	88.55	90.14	74.57	70.60	99.43
5	91.69	67.06	83.33	90.69	88.97	93.58	79.81	96.58	94.21
6	95.39	97.40	81.43	98.70	93.61	94.11	95.84	96.84	97.46
7	100.00	96.15	99.23	96.92	96.92	56.71	94.61	60.44	100.00
8	99.23	99.61	95.92	97.56	93.15	98.69	97.39	99.72	99.97
9	100.00	100.00	100.00	100.00	99.00	18.59	96.00	47.60	100.00
10	83.76	76.61	81.18	80.93	83.22	83.19	48.23	79.85	88.11
11	65.30	70.30	60.9	69.09	69.86	96.26	50.02	87.09	83.70
12	79.05	92.67	73.85	85.47	74.52	83.49	56.36	76.98	88.48
13	99.68	99.63	97.86	99.42	99.27	87.07	99.27	97.15	99.89
14	91.16	99.08	85.10	94.34	91.66	98.22	89.02	98.90	95.92
15	96.94	71.38	92.36	91.00	88.52	81.52	61.41	81.14	97.47
16	98.66	99.46	95.60	99.73	98.13	79.00	99.86	87.35	99.20
OA	82.70	83.18	75.62	81.91	82.77	89.24	64.00	84.78	90.06
AA	91.56	89.41	86.48	89.65	89.10	80.76	76.89	82.89	94.88
κ	0.805	0.810	0.725	0.796	0.806	0.878	0.598	0.828	0.887

Fig. 7 analyzes the effect of matrices' size of **B**, **C**, and **D** on OA. For this image, the OA curve stays stable when r is greater than 30. Thus, all experiments use 40 for matrices' width.

TABLE V
ABLATION ANALYSIS OF THE PROPOSED KR-MLR FOR ALL DATASETS

Method	λ	β_1	β_2	Houston 2018	Xiong'an	Houston 2013	Indian Pines
MLR	×	×	×	–	–	–	–
KR-MLR-1	×	√/×	√/×	–	–	–	–
KR-MLR	×	×	×	65.64	76.98	82.87	80.50
KR-MLR	×	√	×	69.89	79.11	83.98	86.10
KR-MLR	×	×	√	66.10	78.06	87.27	82.63
KR-MLR	×	√	√	70.10	80.10	87.51	87.57
MLR	√	×	×	62.35	76.10	82.04	82.70
KR-MLR-1	√	√/×	√/×	–	–	–	–
KR-MLR	√	×	×	67.75	78.38	82.90	84.60
KR-MLR	√	√	×	75.24	82.48	84.04	89.89
KR-MLR	√	×	√	68.37	79.32	87.31	85.06
KR-MLR	√	√	√	75.49	82.74	88.66	90.06
KR-MLR-2	√	√	√	74.77	81.83	87.10	89.56
KR-MLR-4	√	√	√	73.03	81.05	83.37	89.30

TABLE VI
RUNNING TIME (S) FOR THE PROPOSED KR-MLR AND OTHER ALGORITHMS

image	MLR	SVM	fast-3D-CNN	SSRN	KR-MLR
Houston 2018	86.95	957.64	134.34	4334.67	124.08
Xiong'an	416.08	3576.48	1002.13	20603.43	442.38
Houston 2013	35.69	39.93	47.28	8043.06	61.71
Indian Pines	1.84	2.73	10.11	177.37	12.88

E. Ablation Analysis

KR-MLR provides a joint learning framework to optimize feature and classifier concurrently. The core is Khatri-Rao decomposition and feature learning. Besides, it involves three regularizations parameterized by λ, β_1, and β_2, where β_1 and β_2 are used to introduce discriminant prior and Gaussian prior for factor matrices in Khatri-Rao decomposition and λ is the intrinsic parameter of MLR to avoid ill-condition issue [14]. Next, an ablation study is designed to investigate whether Khatri-Rao decomposition works for KR-MLR especially under HSI classification task, whether feature learning provides performance improvement, and whether all these regularizations are necessary and mutually complemented.

The experiments include four parts.

1) It is to compare the results produced by KR-MLR both with and without Khatri-Rao decomposition.

2) It is to compare the results produced by feature learning-based KR-MLR and the baseline MLR. They are to verify the effectiveness of feature learning especially the importance of Khatri-Rao decomposition in feature learning.

3) It is to test KR-MLR by enabling different regularizations to show their compatibility.

4) It is to evaluate KR-MLR with different Khatri-Rao decomposition orders and illustrate why three-order Khatri-Rao decomposition is used.

The results are reported in Table V. In Table V, KR-MLR-n means n-order Khatri-Rao decomposition is used for feature learning. Specially, KR-MLR-3 is just the proposed KR-MLR and KR-MLR-1 is vector-based feature learning model in (13) without any order Khatri-Rao decomposition.

Several interesting points can be figured out through this ablation study.

1) KR-MLR-1 cannot work for its unbearable computation cost under any conditions. Besides, MLR cannot work without λ since it hits numerical difficulty. In contrast, KR-MLR works successfully even if without any regularizations. It proves that Khatri-Rao decomposition makes the feature learning feasible in computation and avoids the ill-condition issue.

2) When only λ is enabled, it can be found that KR-MLR runs better than MLR. That means Khatri-Rao decomposition-based feature learning can boost the classification performance of KR-MLR. At the same time, it should be noted that the improvement on Houston 2013 is not as large as Houston 2018. It implies that the initial Khatri-Rao decomposition still has large space to grow. Thus, the extra regularizations are developed to further enhance its discriminant ability and convert it into a more effective feature learning tool.

3) As different regularization is gradually added, KR-MLR gets increasing accuracy gain. It confirms that all these regularizations are compatible.

4) For Houston 2018, Xiong'an, and Indian pines, β_1 shows great effect in OA improvement than β_2 while the opposite is true for Houston 2013. The difference may be resulted from the inherent characteristic of datasets themselves. It further proves that both β_1 and β_2 are indispensable.

5) As expected, KR-MLR provides the best OA when all parameter switches are enabled. Under this scenario, the detailed parameter analysis has been discussed in Figs. 8, 12, and 16.

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TABLE VII
CLASSIFICATION RESULTS OF FOUR IMAGES USING UNIFORM PARAMETERS

	Houston 2018	Xiong'an	Houston 2013	Indian Pines
Optimal parameters	75.49	82.74	88.66	90.06
Uniform parameters	73.78	81.88	84.09	87.03

6) KR-MLR outperforms KR-MLR-2 and KR-MLR-4 that explain why three-order Khatri-Rao decomposition is utilized.

F. Computational Cost Analysis

It is necessary to perform some computational complexity analysis for the proposed KR-MLR. KR-MLR involves two loops. For each outer iteration, there are four inner loops. By ignoring the lower order terms, its computational complexity can be written as $O(d_1d_2d_3nr \times \text{iter}2 + (d_1^2 + d_2^2 + d_3^2)nr^2 + (d_1^3 + d_2^3 + d_3^3)r^3)$. Moreover, Table VI reports the running time. In Table VI, KR-MLR and other methods are implemented in python on workstation with Intel(R) Xeon(R) CPU E5-2690 v4 2.60 GHz (28 cores) and 260-GB memory in Linux environment. It should be noted that the choice of such powerful machine is to avoid out-of-memory error caused by fast-3D-CNN and SSRN on the very challengeable and large-scale Xiong'an image. The workstation used here is exactly the minimum-resource machine available in IT storeroom that can run them successfully. For fair comparison, only running times of methods implemented in the same experiment condition are listed. For SVM, in addition to the training and testing times in Table VI, it also takes 40.03, 30.85, 28.60, and 1.72 s for parameter selection for Houston 2018, Xiong'an, Houston 2013, and Indian Pines, respectively. Overall, the running time of KR-MLR is acceptable. Especially, on Xiong'an image, it spends about 440 s for KR-MLR while more than 20 000 s for the benchmark algorithm SSRN. It illustrates the usability of KR-MLR in large-scale dataset.

G. Uniform Parameters

To make KR-MLR easy to use, a uniform parameter combination for λ, β_1, and β_2 is provided. Table VII reports the classification results of four images using $\lambda = 10^6$, $\beta_1 = 10^{-8}$, and $\beta_2 = 10^{-2}$. For comparison, the optimal results are given as well. In Table VII, the uniform parameters, as expected, produce lower accuracies than the optimal parameters. For all that, KR-MLR with uniform parameters is still competitive when compared with the results of others in Tables I–IV even if they are equipped by optimal parameters. It outperforms all compared ones on Houston 2018 and Xiong'an and is second only to IAP on Houston 2013 and GTR on Indian Pines. The experiments of KR-MLR under uniform parameters further show its high performance.

V. CONCLUSION

In this article, KR-MLR is proposed for HSI classification. It combines MLR classifier training and feature learning into a unified framework. Moreover, with Gabor feature as an input, KR-MLR introduces two tailored strategies. One is to convert the feature learning problem into tensor space instead via Khatri-Rao decomposition to make it feasible in computation and greatly reduce the computation cost. Another is to add a local regularization to enhance discriminant ability. The regressors and feature factor matrices are optimized in iterative fashion. KR-MLR yields superior classification accuracy on several HSI datasets, which proves it a competitive and promising classification technique when compared with prior arts.

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REFERENCES

[1] M. Borengasser, W. S. Hungate, and R. Watkins, *Hyperspectral Remote Sensing: Principles and Applications*. Boca Raton, FL, USA: CRC Press, 2008.
[2] C. I. Chang, *Hyperspectral Data Exploitation: Theory and Applications*. New York, NY, USA: Wiley, 2007.
[3] L. G. Olmanson, P. L. Brezonik, and M. E. Bauer, "Airborne hyperspectral remote sensing to assess spatial distribution of water quality characteristics in large rivers: The Mississippi River and its tributaries in Minnesota," *Remote Sens. Environ.*, vol. 130, pp. 254–265, Mar. 2013.
[4] H. M. Dierssen, A. Chlus, and B. Russell, "Hyperspectral discrimination of floating mats of seagrass wrack and the macroalgae Sargassum in coastal waters of Greater Florida Bay using airborne remote sensing," *Remote Sens. Environ.*, vol. 167, pp. 247–258, Sep. 2015.
[5] A. M. Baldridge, S. J. Hook, C. I. Grove, and G. Rivera, "The ASTER spectral library version 2.0," *Remote Sens. Environ.*, vol. 113, no. 4, pp. 711–715, Apr. 2009.
[6] J. Schmedtmann and M. Campagnolo, "Reliable crop identification with satellite imagery in the context of common agriculture policy subsidy control," *Remote Sens.*, vol. 7, no. 7, pp. 9325–9346, Jul. 2015.
[7] G. Camps-Valls, D. Tuia, L. Bruzzone, and J. A. Benediktsson, "Advances in hyperspectral image classification: Earth monitoring with statistical learning methods," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 45–54, Jan. 2013.
[8] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.
[9] J. Richards and X. Jia, *Remote Sensing Digital Image Analysis: An Introduction*. New York, NY, USA: Springer-Verlag, 2006.
[10] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.
[11] W. Li, E. W. Tramel, S. Prasad, and J. E. Fowler, "Nearest regularized subspace for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 477–489, Jan. 2014.
[12] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Hyperspectral image segmentation using a new Bayesian approach with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3947–3960, Oct. 2011.
[13] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4085–4098, Nov. 2010.
[14] J. Li, J. Bioucas-Dias, and A. Plaza, "Spectral-spatial hyperspectral image segmentation using subspace multinomial logistic regression and Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 3, pp. 809–823, 2012.
[15] Y. Qian, M. Ye, and J. Zhou, "Hyperspectral image classification based on structured sparse logistic regression and three-dimensional wavelet texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 2276–2291, Apr. 2013.
[16] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:43:34 UTC from IEEE Xplore. Restrictions apply.

[17] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, doi: 10.1109/LGRS.2020.3043710.

[18] X. Zhang, Y. Liang, C. Li, N. Huyan, L. Jiao, and H. Zhou, "Recursive autoencoders-based unsupervised feature learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1928–1932, Nov. 2017.

[19] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.

[20] W. Sun *et al.*, "A label similarity probability filter for hyperspectral image postclassification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6897–6905, 2021.

[21] J. Peng *et al.*, "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[22] B. Krishnapuram, D. Williams, Y. Xue, A. Hartemink, L. Carin, and M. Figueiredo, "On semi-supervised classification," in *Proc. 18th Annu. Conf. Neural Inf. Process. Syst.*, Vancouver, BC, Canada, 2004, pp. 721–728.

[23] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[24] B. Krishnapuram, L. Carin, M. A. T. Figueiredo, and A. J. Hartemink, "Sparse multinomial logistic regression: Fast algorithms and generalization bounds," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 6, pp. 957–968, Jun. 2005.

[25] W. Sun, K. Ren, X. Meng, C. Xiao, G. Yang, and J. Peng, "A band divide-and-conquer multispectral and hyperspectral image fusion method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2020.3046321.

[26] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3085870.

[27] Q. Shi, X. Tang, T. Yang, R. Liu, and L. Zhang, "Hyperspectral image denoising using a 3-D attention denoising network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10348–10363, Dec. 2021.

[28] D. He, Q. Shi, X. Liu, and Y. Zhong, "Deep subpixel mapping based on semantic information modulated network for urban land use mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10628–10646, Dec. 2021.

[29] H. Guo, Q. Shi, A. Marinoni, B. Du, and L. Zhang, "Deep building footprint update network: A semi-supervised method for updating existing building footprint from bi-temporal remote sensing images," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112589.

[30] K. Bernard, Y. Tarabalka, J. Angulo, J. Chanussot, and J. A. Benediktsson, "Spectral–spatial classification of hyperspectral data based on a stochastic minimum spanning forest approach," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2008–2021, Apr. 2012.

[31] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. A. Chanussot, and J. C. Tilton, "Advances in spectral-spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2012.

[32] G. Mountrakis, J. Im, and C. Ogole, "Support vector machines in remote sensing: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 3, pp. 247–259, 2011.

[33] L.-Z. Huo and P. Tang, "Spectral and spatial classification of hyperspectral data using SVMs and Gabor textures," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Vancouver, BC, Canada, Jul. 2011, pp. 1708–1711.

[34] N. Falco, J. A. Benediktsson, and L. Bruzzone, "Spectral and spatial classification of hyperspectral images based on ICA and reduced morphological attribute profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6223–6240, Nov. 2015.

[35] X. Zhang, Z. Gao, L. Jiao, and H. Zhou, "Multifeature hyperspectral image classification with local and nonlocal spatial information via Markov random field in semantic space," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1409–1424, Mar. 2018.

[36] W. Li and Q. Du, "Gabor-filtering-based nearest regularized subspace for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1012–1022, Apr. 2014.

[37] M. M. Dalla, A. Villa, J. A. Benediktsson, J. Chanussot, and L. Bruzzone, "Classification of hyperspectral images by using extended morphological attribute profiles and independent component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 3, pp. 542–546, May 2011.

[38] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[39] X. Guo, X. Huang, L. Zhang, L. Zhang, A. Plaza, and J. A. Benediktsson, "Support tensor machines for classification of hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3248–3264, Jun. 2016.

[40] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "Generalized tensor regression for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1244–1258, Feb. 2020.

[41] Y. Tarabalka, J. Chanussot, and J. A. Benediktsson, "Segmentation and classification of hyperspectral images using watershed transformation," *Pattern Recognit.*, vol. 43, no. 7, pp. 2367–2379, Jul. 2010.

[42] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Spectral-spatial classification of hyperspectral data using loopy belief propagation and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 2, pp. 844–856, Feb. 2013.

[43] M. Haghighat, S. Zonouz, and M. Abdel-Mottaleb, "CloudID: Trustworthy cloud-based and cross-enterprise biometric identification," *Expert Syst. Appl.*, vol. 42, no. 21, pp. 7905–7916, Nov. 2015.

[44] D. A. Clausi and M. E. Jernigan, "Designing Gabor filters for optimal texture separability," *Pattern Recognit.*, vol. 33, no. 11, pp. 1835–1849, Nov. 2000.

[45] D. Böhning, "Multinomial logistic regression algorithm," *Ann. Inst. Statist. Math.*, vol. 44, no. 1, pp. 197–200, Mar. 1992.

[46] C. C. Chang and C. J. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 27:1–27:27, 2011.

[47] Y. Boykov, O. Veksler, and R. Zabih, "Efficient approximate energy minimization via graph cuts," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 12, pp. 1222–1239, Nov. 2001.

[48] *IRTES CASI-1500 Houston 2018 Dataset*. Accessed: May 2021. [Online]. Available: https://ieee-dataport.org/open-access/2018-ieee-grss-data-fusion-challenge-%E2%80%93-fusion-multispectral-lidar-and-hyperspectral-data#files

[49] *Aerial Hyperspectral Remote Sensing Application Dataset in Xiong'an*. Accessed: May 2021. [Online]. Available: http://www.geodoi.ac.cn/edoi.aspx?Id=1687

[50] *IRTES CASI-1500 Houston 2013 Dataset*. Accessed: May 2021. [Online]. Available: https://machinelearning.ee.uh.edu/2013-ieee-grss-data-fusion-contest

[51] *AVIRIS NW Indiana's Indian Pines 1992 Data Set*. Accessed: Sep. 2020. [Online]. Available: http://cobweb.ecn.purdue.edu/~/MultiSpec/documentation.html



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